Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/09/2026
PI Name:	Garcia-Cardena, Guillermo Ph.D.		
Project Title:	Modeling Anthracycline-triggered Vascular Dysfunction		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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City:	Boston	**State:**	MA
Zip Code:	02115-5727	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	09/27/2022	**End Date:**	09/26/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
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Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Azizgolshani, Hesham Ph.D. (Charles Stark Draper Laboratory Inc) Williams, Corin Ph.D. (Charles Stark Draper Laboratory Inc)		
Grant/Contract No.:	80ARC022CA007		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Recent advances in the development of complex human in vitro models have vastly expanded our potential for biological and drug discovery. Nevertheless, current systems do not have the capability to assess the effects of acute and chronic stressors on cellular systems for long periods of time. This feature is critical to model the effects of certain stressors on relevant target tissues. An important class of such stressors are chemotherapeutic agents currently in use for the treatment of cancer. The primary goal of cancer treatment is to eradicate and prevent the recurrence of cancer. Importantly, achieving this goal has revealed an unintended consequence of therapy: an increased incidence in cardiovascular disease. The acute effects of cancer treatment primarily impact the vascular system, specifically leading to hypertension, vasospasm, and thrombosis. Late effects include long-term toxicities that generally involve chronic cardiovascular changes, including atherosclerosis, valvular heart disease, and conduction system disease. In particular, the use of anthracyclines, a class of chemotherapeutic drugs, is associated with clinical heart failure and other cardiovascular pathologies including hypertension, vascular inflammation, and ischemic heart disease. The mechanisms underlying these adverse effects of anthracyclines on the vasculature have not yet been fully elucidated. In recent years, several studies have reported that anthracycline-treated patients have an impaired flow-mediated vasodilation response, indicating a potential vascular etiology of the adverse cardiovascular effects of this class of drugs. In this project, previously unrecognized actions of the anthracycline, doxorubicin, in vascular endothelial cells recently uncovered by the Garcia-Cardena Laboratory, will provide the basis for establishing a novel conceptual and experimental framework seeking to model the long-term effects of doxorubin in cells of the vascular wall. To this end we will adapt a now well-established, first-of-its-kind high throughput organ-on-chip platform (PREDICT96) developed by the Draper Lab Team to extend the in vitro longevity of vascular tissues (composed of human endothelial, smooth muscle cells, and fibroblasts) to at least 6 months under flow conditions. We will then investigate how doxorubicin acts as an acute and chronic stressor of vascular cells, leading to changes in gene expression, global epigenetic states, and functional phenotypes. In particular, we will focus on the emergence of a senescence, pro-inflammatory phenotype using molecular readouts and cell-based assays to assess how the interactions between vascular endothelial cells and leukocytes evolved in the context of acute and chronic doxorubin treatment. Ultimately, gaining a better understanding of how doxorubicin triggers vascular cell dysfunction should help in the development of new therapeutic interventions to curtail these off-target effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project provides an exceptional opportunity for the development and deployment of a novel generation of Microphysiological Systems designed to support long-term models of the human vasculature. Moreover, these studies should provide the basis for establishing novel conceptual an experimental frameworks to study the effects of chemotherapeutic agents on the human vascular system. Together, these new findings could be very valuable for long-term space missions.
Task Progress:	No reporting received. Please see the Cumulative Bibliography for peer-reviewed papers published with the support of this funding (Ed.,
Bibliography Type:	Description: (Last Updated: 07/09/2026)
Articles in Peer-reviewed Journals	Blazeski A, García-Cardeña G, Kamm RD. "Advancing cardiac organoid engineering through application of biophysical forces." IEEE Rev Biomed Eng. 2025;18:211-30. https://doi.org/10.1109/rbme.2024.3514378 ; PMID: 40030454; PMCID: PMC12146432 , Jan-2025
Articles in Peer-reviewed Journals	Allbritton-King JD, García-Cardeña G. "Endothelial cell dysfunction in cardiac disease: Driver or consequence?" Front Cell Dev Biol. 2023 Oct 25;11:1278166. https://doi.org/10.3389/fcell.2023.1278166 ; PMID: 37965580; PMCID: PMC10642230 , Oct-2023